

Program 1

Introduction to maven and Gradle
Overview of Build Automation tools
Key Differences between maven and gradle
Installation and Setup

⇒ Steps to install maven

1. `sudo apt update`
password we have to give and install
2. `sudo apt install openjdk-17-jdk`
3. `sudo apt install eclipse`
4. `sudo apt install maven` -y
space
5. `mvn -version`
`mvn -version`

Output

Apache Maven 3.6.3

Maven home: /usr/share/maven

Java version: 17.0.18, Vendor: Ubuntu,

runtime: /usr/lib/jvm/java-17-open-amd64

Default locale: en-IN, platform: UTF-8

OS name: linux

version: "6.8.0.90-generic"

arch: "amd64"

family: unix

Steps to Install Gradle ^{Green Cloud}

1. `wget -c https://services.gradle.org/distributions/gradle-7.4.2-bin.zip`
2. `sudo apt install unzip`
3. `sudo unzip -d /opt /gradle/tmp/gradle-*.zip`
4. `sudo ls /opt / gradle`
5. `sudo nano /etc/profile.d / gradle.sh`
6. `sudo nano /etc/profile.d / gradle.sh`
7. `sudo chmod +x /etc/profile.d / gradle.sh`
8. `sudo source /etc/profile.d / gradle.sh`
9. `source /etc/profile.d / gradle.sh`
10. `gradle -v`

output

Gradle 7.4.2

Build time : 2022-03-31

Revision : 504073681146c7cc26469cbc

Kotlin : 1.5.3.1

Groovy : 3.0.9

Ant : ~~Apache~~

JVM : 17.0.13

OS : Linux 6.8.08-90

11/2/22

② Creating a Maven project, Understanding the Pom File Dependency Management and Plugin. *Groupid.

To run and complete we use plugin.

* Groupid, * Artifact Id,

* archetype in Maven

Pom

Project object maven

Step 1:

open eclipse

home / student / eclipse

— workspace

↓
Launch

In eclipse tool Go to File - New - Maven Project

☒ default workspace

Create Groupid :- pom. program2

Artifact Id :- program2 - Example.jar.

java version must be 17

Sudo update alternatives --config java
" " " " " " " " " " " "

After the above procedure is done
Select goal as package

↓
Click on Run

↓
Automatically console screen opens finally displays

Step 3:- In eclipse left top corner

☐ Src/main/java (double click)

↓
☐ Com. program2.maven.program2

↓
- Click on App.java

Step 4:- Finally run the program

☐ Run App.java

↓
Display the final output screen
"Hello world"

Eclipse → launch →

File → New → maven file

↓
Group ID → unique

Artifact ID

↓
Right click Run as maven build next
→ It will build the file name

⑧ Working with Gradle Setting up a Gradle project Understanding Build scripts (Groovy and Kotlin), Dependency Management as Task Automation.

Step 1:- Go to terminal

Step 2:- Then make a new directory
mkdir username
cd username

Step 3:- If we want to create a project in Gradle then we use Command

\$ gradle init /* create a project by using gradle */

Step 4:- then select type of project to generate

1. basic /* want to create basic application using gradle */

2. Application /* want to create application using gradle */

3. Library /* want to create library using gradle */

4. Gradle plugin /* want to create one plugin using gradle */

/* plugin is used to extend the functionality of a project. plugins add new false and

features making it easier to build
list and manage dependency.

Step 5: type 2 (application)

Step 6: Next have to select for Implementation
language

1. C++
2. Groovy
3. Java
4. Kotlin
5. Scala
6. Swift

After type 3, (Java)

Split functionality across multiple

Subprojects ?

1. no - only one application project
2. yes - application and library projects

Enter selection:

type 1 (no)

Step 7: Select build script DSL

1. Groovy
2. Kotlin

Enter selection

type 1 (Groovy)

Generate build using new API and

behaviors [yes, no] : no type no

Step 8: Select test framework

1. JUnit 4

2. TestNG

3. Spock

4. JUnit Jupiter

Enter selection : 1 (JUnit 4)

Step 9: Add project name, source package

Project name: gradle-test

Source package: gradle-test

Project name: gradle-test
Source package: gradle-test } no space.

mkdir pushpa

cd pushpa

gradle init

select application

type 2

Select lang

java

type 3

For ~~select~~ Functionality

type no

Select 1

Build script

1. Groovy
2. Kotlin

Select 1

Frame work

1. JUnit

Project name

Source name

any names without keyword

Build successfully

Program - 5

4) ~~Practical Exercise~~ Build and Run a Java Application with maven. Migrate the same Application to Gradle.

Step 1:- Follow program 2 steps as it is from first to last

User Admin
Pass: Siet@00

⑤ Introduction to Jenkins what is Jenkins:
Installing Jenkins on local or Cloud environment
Configuration.

Step 1:- To check Java is Install (or) not then type
Java --version

Step 2:- Go to the Browser In Google

① Search download Jenkins in ubuntu

② Click on link download and deployment

③ In this page down we get two list

+ Download Jenkins 2.3.7.5.1.2B.5

↑
Select

Click ubuntu/Debian

Then commands to display then just copy and past to the terminal one by one.

1. Sudo - wget -O /usr/share. ← ←

2. Echo "deb [signed-by = /usr --- ←

3. Sudo apt-get update ←

4. Sudo apt -get install Jenkins ←
5. We can enable the Jenkins service to start at boot with the command.

Sudo systemctl Enable Jenkins ←

6. We can start the Jenkins services with the command

Sudo systemctl start Jenkins ←

7. We can check the status of the Jenkins service using the command

Sudo systemctl status Jenkins ←

Step 3:- Now setup the post Installation

8. Go to the browser In url type
localhost :8080 ←

default port user for Jenkins.
In that copy the path & paste

9. In the terminal to get password (In terminal)

Sudo more [usr/lib] jenkins/Secrets/
Initial Admin password ←

It displays password copy this password.

10. Paste above passwords to Jenkins page window
~~then we will get~~ [displayed in browser]
and click on Continue.

11. Now we will get customize Jenkins page in
that we will get.

- ① Install Suggest plugins
- ② Select plugins to install

click on ① is install suggested plugins
wait for few min it will install all plugin

③ next it will open sign up page automatically
once all plugin is get install as below.

first Admin user

Username → ~~Admin~~ Pushpa

Password → Reddy@123

Confirm password → Reddy@123

Full name → Pushpanjali

Email address → pushpanjali.sujatha@gmail.com

Press Save and Continue.

④ Then we will get Jenkins url

http://localhost:8080

click on Save and Finish

Then we will get page as
Jenkins is ready

Click start using Jenkins

To login: url: localhost:8080

Program -6

Continuous Integration with Jenkins, setting up a CI pipeline, Integrating Jenkins with maven, Gradle running Automated Builds and Tests.

First

In terminal

- ① `mkdir pushpa` // create one directory
`cd pushpa`
- ② `mvn archetype:generate -DgroupId=com.example`
`-DartifactId=my-app -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false`
- ③ `ls`
`my-app`
- ④ `cd my-app`
- ⑤ `mvn clean install`
- ⑥ `git init`
- ⑦ `git add`
- ⑧ `git commit -m "pom Example"`
- ⑨ `git branch -m main`
- ⑩ `git remote add origin (url github copy)`
in git and paste here
- ⑪ `git push -u origin main`

In Jenkins

localhost : 8080

login : Pushpa

Reddy@123

First we must add SSH key to Jenkins
Credentials

SSH key what copy In ①

1. Dash Board
2. Manage Jenkins (left corner in Dashboard)
3. Select the appropriate domain (global)
4. Add new credentials

click on add Credentials, under that
kind → SSH username with password.

scope → Select Global (Jenkins, nodes, items,
all child items etc.)

username : Enter git

click private key

0 Enter direct
key

Paste the SSH code → ① permissions

press create.

In Jenkins

Dashboard → New Item → name (secret123) →
free style → ok →

Can Configure:

git

- ① Repository clnl
Paste Github repository code ssh
- ② Branches to build
Branch specifier
*/main

③ Environment

- ✓ Delete workspace before build start
- ✓ Add time stands to console o/p.
- ✓ Terminate a build if it's stuck
Time out min

20

Apply Save

Run (Build Also)

Step 2:- Set up an SSH key pair

The ansible control node uses SSH to connect to hosts, Generate an SSH key pair for the ansible user by executing the following steps

1) Enter the command below the ansible control node command line

`ssh-keygen`

press `<enter>` to display command line

i.e., `ssh@localhost`

Step 3: Configure ansible host

To know remote-host to next command type

`hostname -I`

It display same remote host

i.e. 10.100.50.90

1. use the following `ssh-copyid` command on the control node to copy the public key to a host

`ssh-copyid pumpa @ 10.100.50.90.`

ask password

⇒ we need to check ansible there or not
open `ssh-server`

For installation

`sudo apt install`

`if`

`sudo apt install update.`

`openssh-server`

If

connection
fails

Step 4: Install Ansible

Use the APT package manager to install the ansible package on the control node

1. Ensure the package index is up to date
Sudo apt install update

2. Install Ansible on ubuntu with the following
Command

Sudo apt install ansible - y

Step 5:- Verify the installation
ansible --version

Step 6:- Set up the inventory file

Follow the steps below to create an inventory
file on the control node

1) Create the ansible subdirectory in the etc-
directory

Sudo mkdir -p /etc/ansible

2) Use a text editor such as nano to
create a file named host

Sudo nano /etc/ansible/hosts

3) Add localhost that control node will manage
use the following format [local]

localhost ansible_connection=local

4) Save the file and exit

5) Enter the Command below to check items in
the inventory

ansible-inventory --list - y

Step 7 : Test the connection

Sudo ansible all -m ping

Note - for success

Sudo systemctl start ssh

Sudo systemctl status ssh

Set the path

[local]

localhost

ansible_connection = local

Program-8

⑧ Practical Exercise :- Set-up a Jenkins pipeline for maven project, use Ansible to deploy artifacts generated by Jenkins.

Step 1 :- In Terminal

1) Install Ansible

Sudo apt install ansible -y ← ansible --

2) Copy path
pwd ← present working directory
copy the path → ①

In Jenkins

localhost : 8080 ←

username : Pushpa

Password : Pushpa@123

Step 1 :- Install Ansible tools

Dashboard → manage Jenkins → plugin →
Ansible plugin

Install : Ansible

Ansible Tower

Step 2 :- To set the path

Dashboard → manage Jenkins → tool →
Scroll down we get section click on that
it will ask name → ansible

Path → for the path

Path → paste the path which is copy from the terminal def ① my be / home / student

Apply → Save

③ Run pipeline

Dashboard → New Item → project name (pipeline)

→ select pipeline → press ok

Scroll down don't do anything

In pipeline

pipeline script

copy and paste the script it is in my mail

so
Apply : Save

Run.

Pipeline Script

pipeline {

agent any

stages {

stage ('Build') {

steps {

echo 'Building' --

}

stage ('Test') {

steps {

echo 'Testing' --

}

}

Stage ('Deploy') {

Steps { echo 'Deploying...' }

}

8 (2)

Jenkins declarative pipelines used to automate CI/CD (Continuous Integration and Continuous Deployment) tasks.

Structure

* pipeline { -- }

This defines the start of a declarative pipeline in Jenkins

* agent any

This tells Jenkins to run the pipeline on any available agent (or node) same of the agents in Jenkins.

1) Build-in (master/Controller Agent)

2) Static Agent

3) Dynamic Agent

5) Label Based Agent

* Stages { -- }

A block containing all the stages of the pipeline

Stage Details

1) Stage ('Build')
used to generate source code etc --

2) Stage ('Test')
usually for running automated test like
unit test Integration test.

3) Stage ('Deploy')
usually for deploying the build app
to a server (environment)